

6

$$\begin{aligned}
 & \frac{1}{4}(\|f + g\|^2 - \|f - g\|^2 - i\|f + ig\|^2 + i\|f - ig\|^2) \\
 & \frac{1}{4}(\langle f + g, f + g \rangle - \langle f - g, f - g \rangle - i\langle f + ig, f + ig \rangle + i\langle f - ig, f - ig \rangle) \\
 & \frac{1}{4}(\|f\|^2 + \|g\|^2 + 2\operatorname{Re}(\langle f, g \rangle) - \|f\|^2 - \|g\|^2 + 2\operatorname{Re}(\langle f, g \rangle) + \\
 & i(-\|f\|^2 + \|g\|^2 + 2\operatorname{Im}(\langle f, g \rangle) + \|f\|^2 - \|g\|^2 + 2\operatorname{Im}(\langle f, g \rangle))) \\
 & \frac{1}{4}(4\operatorname{Re}(\langle f, g \rangle) + 4\operatorname{Im}(\langle f, g \rangle)) \\
 & = \langle f, g \rangle
 \end{aligned}$$

7 Let $\mathcal{H} = L^2([0, \infty), \mathcal{B}, \mu)$. We claim that the function $f : (0, \infty) \rightarrow \mathcal{H}$ given by $f(t) = 1_{[0, t]}$ is continuous and has the property for all $t \in (0, \infty)$ that for all $r < s < t$ we have that $\langle f(t) - f(s), f(s) - f(r) \rangle = 0$. Note that f is continuous since for a given $\epsilon > 0$ we have that if $|t - s| < \epsilon^2$ with WLOG $t > s$ then

$$\|f(t) - f(s)\| = \sqrt{\int_{(0, \infty)} |1_{[0, t]} - 1_{[0, s]}|^2 d\mu} = \sqrt{\int_{(0, \infty)} |1_{(s, t]}|^2 d\mu} = \sqrt{\epsilon^2} = \epsilon.$$

Furthermore the self orthogonality comes from the fact that

$$\begin{aligned}
 \langle f(t) - f(s), f(s) - f(r) \rangle &= \langle f(t), f(s) \rangle + \langle f(s), f(r) \rangle - \langle f(s), f(s) \rangle - \langle f(t), f(r) \rangle \\
 &= s + r - s - r \\
 &= 0
 \end{aligned}$$

For a general Hilbert space we have that:

8

9 Let $\{f_n\}_{n \in \mathbb{N}}$ be a weakly convergent sequence in a Hilbert space $\mathcal{H}$ with limit f . We want to show that $\lim \|f - f_n\| = 0$ iff $\lim \|f_n\| = \|f\|$.

- $\Rightarrow$ Suppose that $\lim \|f - f_n\| = 0$. Then we have that

$$\lim \|f\|^2 + \|f_n\|^2 - 2\operatorname{Re}(\langle f, f_n \rangle) = 0$$

Note by the weak convergence of f_n we have that $\lim \|f\|^2 + \|f_n\|^2 = 2\|f\|^2$, therefore $\lim \|f_n\|^2 = \|f\|^2$, which implies our desired result

- $\Leftarrow$ Suppose that $\lim \|f_n\| = \|f\|$. Then we have by the parallelogram law that

$$\begin{aligned}
 \lim \|f - f_n\|^2 &= \lim 2\|f\|^2 + 2\|f_n\|^2 - \|f + f_n\|^2 \\
 &= \lim 4\|f\|^2 - \|f\|^2 - \|f_n\|^2 - 2\operatorname{Re} \langle f, f_n \rangle \\
 &= 4\|f\|^2 - \|f\|^2 - \|f\|^2 - 2\|f\|^2 \\
 &= 0
 \end{aligned}$$

Note that the last line is true because of the weak convergence of f_n . Therefore $\lim \|f - f_n\| = 0$.

- 10 Let μ be the Borel measure on $[0, 1]$, C be the set of real valued measurable functions f on $[0, 1]$ such that $\int_{[-1,1]} f^2 d\mu = 1$, $\int_{[-1,1]} x^j f d\mu = 0$ for $j \in \{0, 1, 2\}$. The maximizing value of $\int_{[-1,1]} x^3 f(x) d\mu$ is $\frac{2\sqrt{2}}{5\sqrt{7}}$, however I don't know how to prove this. I believe if I applied gram-schmidt then there would've been a point in which the inner product was taken between computable polynomials, however I did not have enough time to do this.